

Sampling of analog signals

I.1. Analog and Digital Signals

Signals

- ▶ Signal = a measurable quantity which varies in time, space or some other variable
- ▶ Examples:
 - ▶ a voltage which varies in time (1D voltage signal)
 - ▶ sound pressure which varies in time (sound signal)
 - ▶ intensity of light which varies across a photo (2D image)
- ▶ Represented as a mathematical function, e.g. $v(t)$.

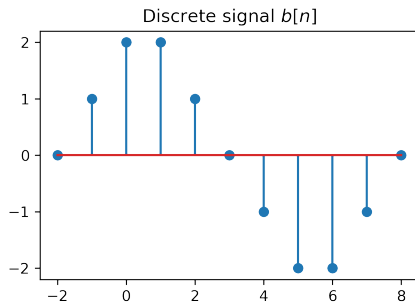
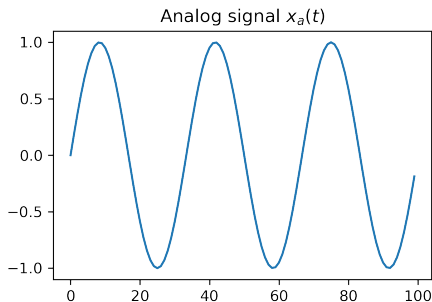
- ▶ Glossary:
 - ▶ “e.g.” = “*exempli gratia*” (lat.) = “for example” (eng.) = “de exemplu” (rom.)
 - ▶ “i.e.” = “*id est*” (lat) = “that is” (eng.) = “adică” (rom.)

Signal dimension

- ▶ **Unidimensional** (1D) signal = a function of a single variable
 - ▶ Example: a voltage signal $v(t)$ only varies in time.
- ▶ **Multidimensional** (2D, 3D ... M-D) signal = a function of multiple variables
 - ▶ Example: intensity of a grayscale image $I(x, y)$ across the surface of a photo
- ▶ In these lectures we consider only 1D signals, but the theory is similar

Continuous and discrete signals

- ▶ Continuous (analog) signal = function of a continuous variable
 - ▶ Signal has a value for every possible value of the variable in the defined range
 - ▶ The variable may be defined only in a certain range (e.g. $t \in [0, 100]$), but it is a continuous range
- ▶ Discrete signal = function of a discrete variable
 - ▶ Signal has values only at certain discrete values (*samples*)
 - ▶ Indexed with integer numbers: $x[-1]$, $x[0]$, $x[1]$ etc.
 - ▶ Outside the samples, the signal is **not defined**



Notation

- ▶ We use the following notation:
- ▶ Continuous signal
 - ▶ Has **round parentheses**, e.g. $x_a(t)$
 - ▶ Sometimes has the a subscript
 - ▶ The variable is usually t (time)
 - ▶ $x_a(2.3)$ = the value of the signal $x_a(t)$ at $t = 2.3$
- ▶ Discrete signal
 - ▶ Has **square brackets**, e.g. $x[n]$
 - ▶ The variables are denoted as n or k (suggest natural numbers)
 - ▶ $x[3]$ = the value of the signal $x[n]$ for $n = 3$
 - ▶ $x[1.5]$ = does not exist

Continuous, sampled and digital signals

- ▶ **Analog signal:** continuous time, continuous amplitude
 - ▶ e.g. a microphone voltage $x_a(t)$
- ▶ **Sampled signal:** discrete time, continuous amplitude
 - ▶ sampling selects the values $x_a(nT_s)$
- ▶ **Digital signal:** discrete time, discrete amplitude
 - ▶ quantization rounds every sample to one of a finite number of levels
- ▶ **Sampling discretizes time; quantization discretizes amplitude**

Periodic signals

- ▶ A signal is **periodic** if the values repeat themselves after a certain time (**fundamental period**)
- ▶ Continuous signals:
 - ▶ Periodic: $x_a(t) = x_a(t + T), \forall t$, where $T > 0$
 - ▶ T is usually measured in seconds (or some other unit)
- ▶ Discrete signals:
 - ▶ Periodic: $x[n] = x[n + N], \forall n$, where $N > 0$
 - ▶ N **has no unit**, because it is just a number

Discrete frequency

- ▶ Harmonic signals have a frequency f :

$$x(t) = 2 \cdot \cos(2\pi \cdot 400 \cdot t + \frac{\pi}{3})$$

$$x[n] = 5 \cdot \sin(2\pi \cdot 0.12 \cdot n + \frac{\pi}{2})$$

- ▶ Pulsation $\omega = 2 \pi \cdot \text{frequency}$

- ▶ Continuous signals:

- ▶ $T = \frac{1}{f}$ is usually measured in seconds (or some other unit)

- ▶ $F = \frac{1}{T}$ is measured in $\text{Hz} = \frac{1}{s}$ (Hertz)

- ▶ Discrete signals:

- ▶ N **has no unit**, because it is just a number

- ▶ f **has no unit** also

Periodicity of a discrete sinusoid

- ▶ Consider

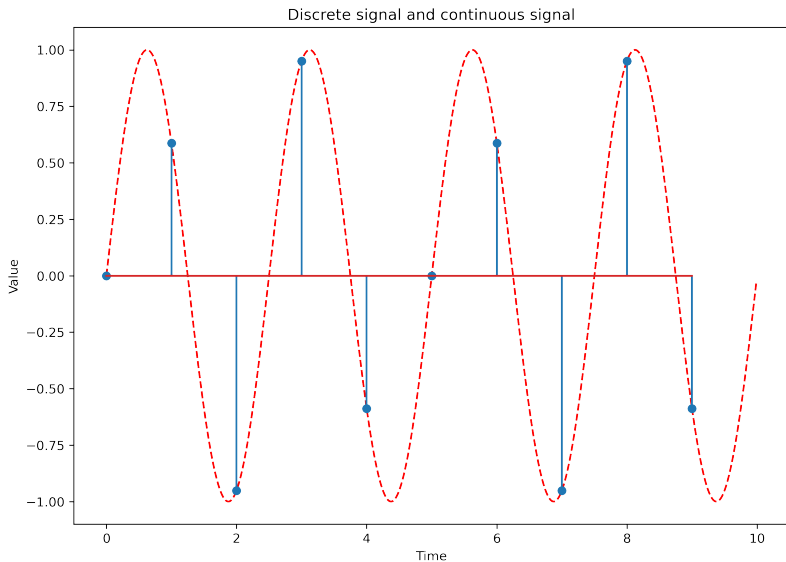
$$x[n] = \cos(2\pi f n)$$

- ▶ It is periodic **if and only if** f is rational
- ▶ If $f = \frac{p}{q}$ is in lowest terms, its fundamental period is

$$N = q$$

- ▶ Example: $f = \frac{3}{8}$ gives $N = 8$
- ▶ Contrast: $f = \frac{\sqrt{2}}{10}$ is irrational, so no finite period exists

Period and frequency



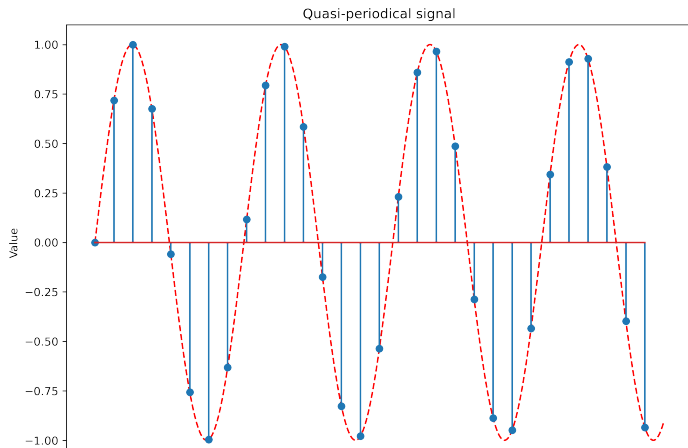
► Discrete signal period $N = 5$, frequency $f = 0.4$

Period and frequency

- ▶ Quasi-periodic signals:

$$x[n] = \sin(0.8 \cdot n)$$

- ▶ Frequency = irrational number, period = does not exist, but the signal is “almost periodic”



Checkpoint: periodicity

- ▶ For $x[n] = \cos(2\pi fn)$, decide whether it is periodic:
 - ▶ $f = \frac{5}{12}$
 - ▶ $f = \frac{\sqrt{3}}{8}$
- ▶ **Answer:** the first signal is periodic with $N = 12$; the second is not periodic
- ▶ Reducing $f = \frac{p}{q}$ to lowest terms is essential

Domain of definition

- ▶ **Finite-length** discrete signals $x[n]$:
 - ▶ have only a certain number N of samples (e.g. for $n = 0, 1, \dots, N-1$)
 - ▶ they are not defined outside these samples
 - ▶ can be represented as a **vector** of numbers (e.g. like in Matlab, C)
- ▶ **Infinite-length** discrete signals $x[n]$:
 - ▶ e.g. defined for $n = \dots, -2, -1, 0, 1, 2, \dots$

Vector space of signals

- ▶ All signals of a certain length N form a **vector space**
- ▶ In mathematics, a vector space = a set V of elements $\{v\}$ (called “vectors’ ’) such that:
 - ▶ the sum of any two elements from V is still a member of V
 - ▶ any vector from V multiplied by a constant is still a member of V
- ▶ These properties can easily be verified for signals

I.2. Sampling

Sampling

- ▶ Sampling = Taking the values from an analog signal at certain discrete moments of time, usually periodic
- ▶ Distance between two samples = **sampling period** T_s
- ▶ **Sampling frequency** $F_s = \frac{1}{T_s}$
- ▶ Why sampling?
 - ▶ Converts continuous signals to discrete
 - ▶ Processing of continuous signals is expensive
 - ▶ Processing of discrete signals is cheap (digital devices)
 - ▶ Sometimes nothing is lost due to sampling

From the physical world to digital processing

phenomenon / sensor $\rightarrow$ anti-alias filter $\rightarrow$ ADC

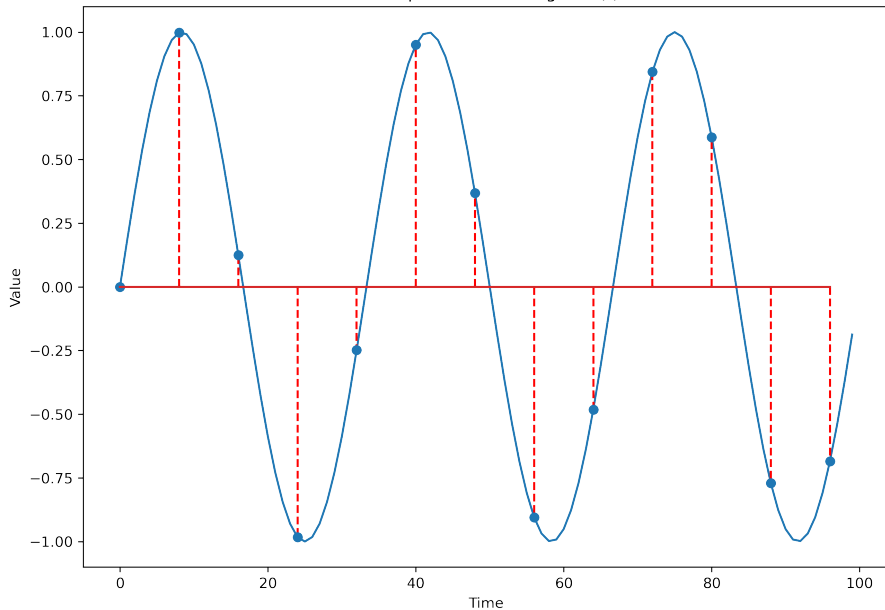
ADC $\rightarrow$ digital processing / storage $\rightarrow$ DAC

DAC $\rightarrow$ reconstruction filter $\rightarrow$ output

- ▶ ADC: **sampling** $\rightarrow$ **quantization** $\rightarrow$ **coding**
- ▶ Filters act in continuous time, before the ADC and after the DAC

Graphical example

A sample sinusoidal signal $v(t)$



Sampling equation

- ▶ Mathematically, it is described by **the sampling equation**, which uses the Dirac delta function $\delta(t)$:

$$x[n] = \int_{-\infty}^{+\infty} x_a(t) \delta(t - nT_s) dt$$

- ▶ For continuous signals, this is equal to the value of the signal at $t = nT_s$:

$$x[n] = x_a(n \cdot T_s)$$

- ▶ Periodic sampling produces a discrete signal $x[n]$ from a continuous signal $x_a(t)$

Sampling of harmonic signals

- ▶ Let's sample a cosine: $x_a(t) = \cos(2\pi Ft)$

$$\begin{aligned}x[n] &= x_a(nT_s) \\&= \cos(2\pi F n T_s) \\&= \cos(2\pi F n \frac{1}{F_s}) \\&= \cos(2\pi \underbrace{\frac{F}{F_s}}_f n) \\&= \cos(2\pi f n)\end{aligned}$$

- ▶ Same for sine instead of cosine

Discrete frequency is relative

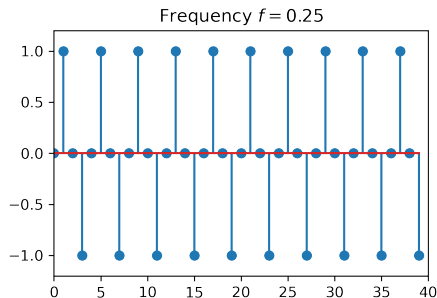
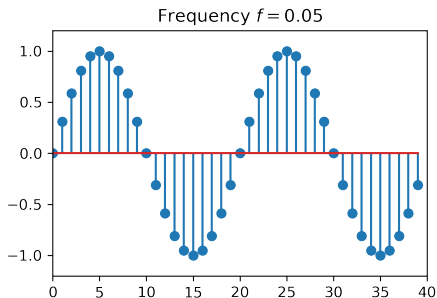
- ▶ Sampling a continuous cosine produces a discrete cosine with **discrete frequency**:

$$f = \frac{F}{F_s}$$

- ▶ Discrete frequency should be understood as a value **relative to the sampling frequency**
- ▶ Example: $f = \frac{1}{4}$ means “coming from an analog frequency F which was $\frac{1}{4}$ of the sampling frequency”
 - ▶ it could have been a 100Hz signal sampled with 400Hz
 - ▶ it could also have been a 3MHz signal sampled with 12MHz

False friends

- **Note:** A discrete sinusoidal signal might not *look* sinusoidal, when its frequency is high (close to $\frac{1}{2}$).



Sampling theorem (Nyquist-Shannon)

The Nyquist-Shannon sampling theorem:

- ▶ If a signal $x_a(t)$, which is band-limited with maximum frequency F_{max} , is sampled with a sampling frequency

$$F_s > 2F_{max},$$

then it can be perfectly reconstructed from its samples using the formula:

$$x_a(t) = \sum_{n=-\infty}^{+\infty} x[n] \cdot \frac{\sin(\pi(F_s t - n))}{\pi(F_s t - n)}.$$

Comments on the sampling theorem

- ▶ All the information in the original signal is contained in the samples, provided the sampling frequency is high enough
- ▶ It is much easier to process discrete samples instead of analog signals (e.g. using Matlab instead of capacitors :))
- ▶ Sampling with $F_s > 2F_{max}$ makes every signal frequency strictly smaller than $1/2$:

$$f = \frac{F}{F_s} \leq \frac{F_{max}}{F_s} < \frac{1}{2}$$

Example of the sampling theorem in action

Sampling theorem in action:

- ▶ Humans can only hear sounds up to $\sim 20\text{kHz}$
- ▶ Sampling rates higher than 40kHz are enough for audible frequencies
 - ▶ Standardized for CD-Audio: 44100Hz

Aliasing

- ▶ <http://www.dictionary.com/browse/alias>:
 - ▶ “alias”: a false name used to conceal one’s identity; an assumed name
- ▶ What happens when the sampling frequency is not high enough?
- ▶ Example: $F = 600Hz$ sampled with $F_s = 1000Hz$

$$\begin{aligned}x[n] &= x_a(nT_s) \\&= \cos(2\pi 600nT_s) \\&= \cos(2\pi 600n \frac{1}{1000}) \\&= \cos(2\pi \underbrace{\frac{6}{10}}_f n)\end{aligned}$$

- ▶ Bad sign: We get a frequency larger than $f_{max} = \frac{1}{2}$

Funny things with $\cos()$ and $\sin()$

- ▶ Discrete $\cos()$ and $\sin()$ have funny properties
- ▶ They are **the same** when adding an integer to the frequency:

$$\cos(2\pi(f+k)n) = \cos(2\pi fn + 2\pi kn) = \cos(2\pi fn)$$

- ▶ So all these discrete frequencies are identical:

$$f = \dots = -1.4 = -0.4 = 0.6 = 1.6 = 2.6 = 3.6 = \dots$$

- ▶ In addition, negative frequencies can be turned into positive:

$$\cos(2\pi(-f)n) = \cos(2\pi fn)$$

$$\sin(2\pi(-f)n) = -\sin(2\pi fn)$$

Aliasing:

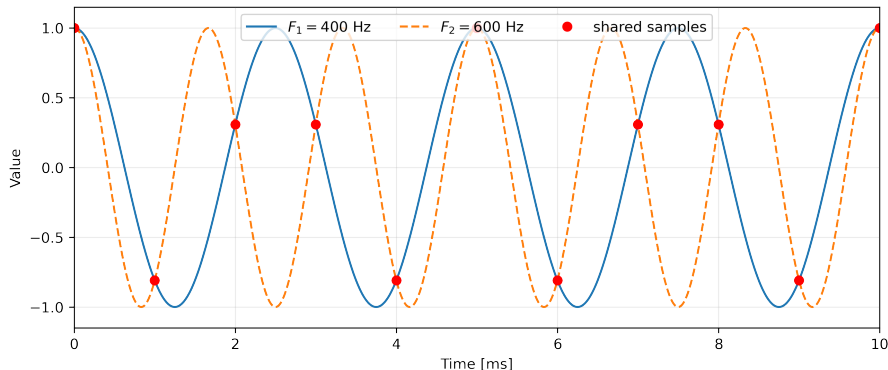
- ▶ Every discrete frequency f outside the interval $[-\frac{1}{2}, \frac{1}{2})$ is **identical** (an “alias”) with a frequency from this interval $f_{alias} \in [-\frac{1}{2}, \frac{1}{2})$
- ▶ Just add or subtract 1's to f until the result is in $[-\frac{1}{2}, \frac{1}{2})$
- ▶ The interval $[-\frac{1}{2}, \frac{1}{2})$ is called the **Nyquist interval** or the **principal interval**

Frequency limits

- ▶ For continuous signals, F can go to ∞
 - ▶ Because period T can be $T \rightarrow 0$
- ▶ For discrete signals, we consider the **largest frequency** to be $f_{max} = \frac{1}{2}$, in the sense that all frequencies f can be reduced to a frequency $\in [-\frac{1}{2}, \frac{1}{2})$

One set of samples, two analog frequencies

- ▶ Let $F_s = 1000$ Hz, $F_1 = 400$ Hz and $F_2 = 600$ Hz
- ▶ Because $0.6 \equiv -0.4 \pmod{1}$, the two cosines have the same samples



The problem of aliasing

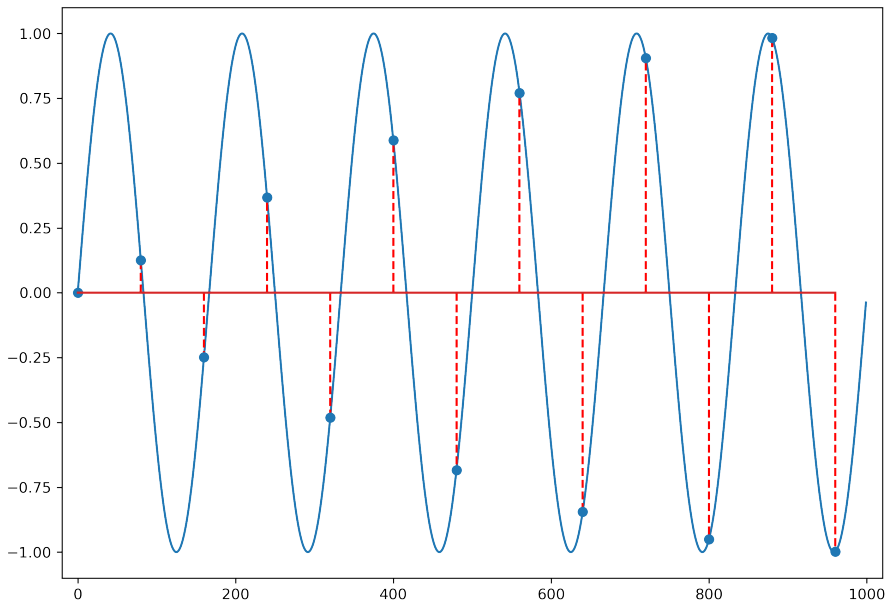
- ▶ Sampling different signals can lead to exactly same samples
- ▶ Problem: how to know from what signal did the samples come from? Impossible.
- ▶ Example:

- ▶ all these discrete frequencies are identical:

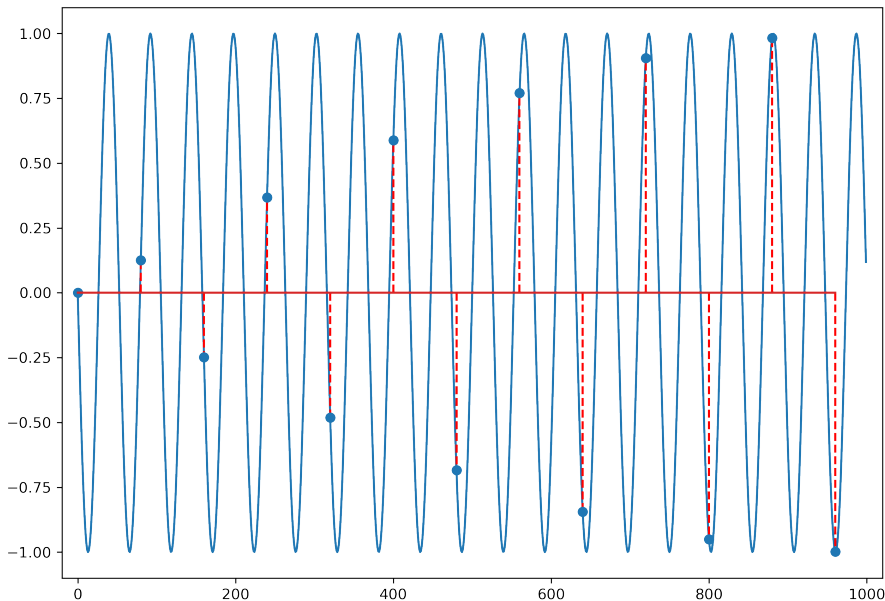
$$f = -0.4 = 0.6 = 1.6 = \dots$$

- ▶ so if $F_s = 1000\text{Hz}$, the original signal could have been any frequency F out of: 600Hz or 1600Hz or ...
 - ▶ Exercise: check some of these

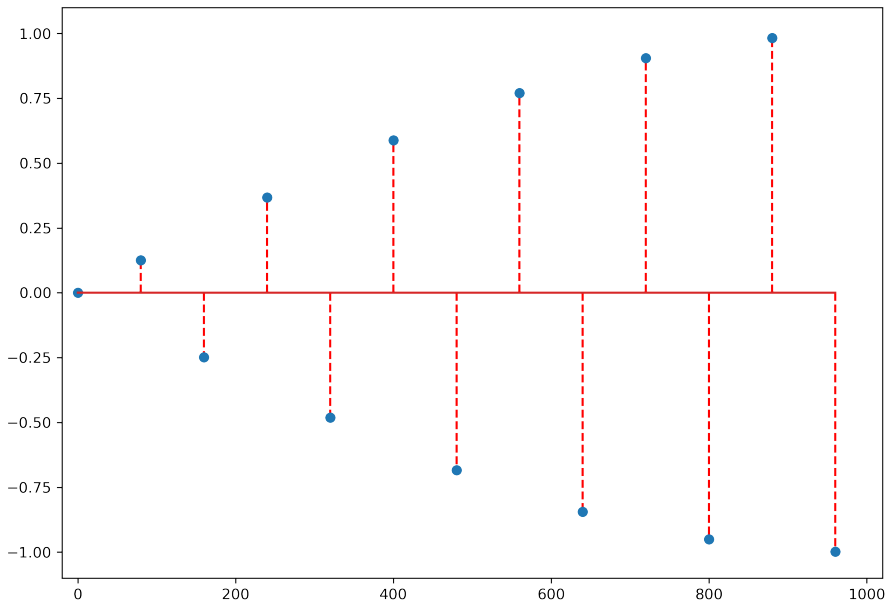
Aliasing example - low frequency signal



Aliasing example - high frequency signal, same samples



Aliasing example - samples only



Checkpoint: sampling and aliasing

- ▶ A 800 Hz cosine is sampled at 1000 Hz. Which cosine in the principal interval has the same samples?

- ▶ Sampling according to Shannon theorem guarantees no aliasing:

$$F_s > 2F_{max} \Rightarrow f = \frac{F}{F_s} < \frac{1}{2}$$

- ▶ It is better remove from the signal all the frequencies larger than $\frac{F_s}{2}$, which will not be sampled correctly, otherwise they will create a false frequency and bring confusion
- ▶ **Anti-alias filter:** a low-pass filter situated before a sampling circuit, rejecting all frequencies $F \geq \frac{F_s}{2}$ from the signal before sampling
 - ▶ Standard practice in the design of processing systems

Ideal signal reconstruction from samples

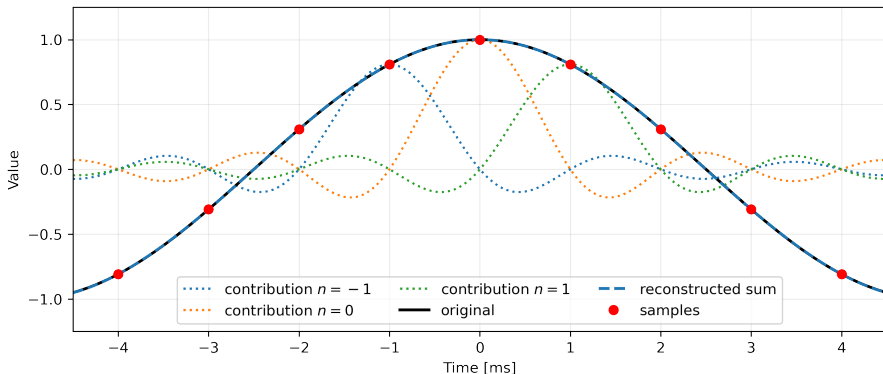
- ▶ Reconstruction = opposite of sampling, producing a continuous signal from a discrete one
- ▶ For harmonic signals ($\sin$, $\cos$), the ideal reconstruction equation amounts with the inverse variable change $n = \frac{t}{T_s} = t \cdot F_s$:

$$x_r(t) = x\left[\frac{t}{T_s}\right] = x[t \cdot F_s]$$

- ▶ A discrete frequency f becomes $F = f \cdot F_s$

Sinc interpolation: ideal reconstruction

- ▶ Example: $F = 100$ Hz sampled at $F_s = 1000$ Hz, safely inside the Nyquist interval
- ▶ We use 61 samples and show only the interior; shifted sinc terms sum through every sample value



Reconstruction and aliasing

- ▶ What value to use for f ?
 - ▶ we know $f = f + 1 = f + 2 = \dots$, which one to use?
- ▶ The reconstruction assumes all f are in the interval $[-\frac{1}{2}, \frac{1}{2})$
 - ▶ apply reconstruction equation
 - ▶ the resulting signal has all frequencies $\frac{F_s}{2} \leq F < \frac{F_s}{2}$ ($\frac{F_s}{2}$ = “the Nyquist frequency”)
- ▶ **In exercises:** Always bring f in the interval $[-\frac{1}{2}, \frac{1}{2})$ before reconstruction
- ▶ Reconstruction always produces signals with frequencies in $[-\frac{F_s}{2}, \frac{F_s}{2})$
 - ▶ Only signals or components sampled according to the sampling theorem will be reconstructed identically
 - ▶ Any other components are replaced with their aliased counterparts

A/D and D/A conversion

- ▶ Sampling + quantization + coding is usually done by an **Analog to Digital Converter (ADC)**
 - ▶ It takes an analog signal and produces a sequence of binary-coded values
- ▶ Reconstructing an analog signal from numeric samples is done by a **Digital to Analog Converter (DAC)**
 - ▶ Usually the reconstruction is not based on sampling theorem equation, which is too complicated, but with simpler empirical solutions
- ▶ You have ADCs and DACs for any In or Out audio jack (phone, computer etc)

Signal quantization and coding

- ▶ In practice, the amplitudes of the samples are converted to binary representation
- ▶ Because of this, the amplitudes are rounded to fixed levels, e.g. 8-bit values (256 distinct levels) , 16-bit values (65536).
- ▶ This “rounding” is known as **quantization**
- ▶ The “rounding error” is known as **quantization error**
- ▶ Converting the value to binary form is known as **coding**
- ▶ ADCs handle sampling, quantization and coding simultaneously

Sampling: formula summary

- ▶ Sampling clock: $F_s = \frac{1}{T_s}$
- ▶ Sampling equation: $x[n] = x_a(nT_s)$
- ▶ Relative frequency: $f = \frac{F}{F_s}$
- ▶ Periodicity: $f = \frac{p}{q}$ in lowest terms $\iff N = q$; irrational f is not periodic
- ▶ Nyquist–Shannon condition: $F_s > 2F_{max}$
- ▶ Frequency equivalence: $f \equiv f + k \pmod{1}$, $k \in \mathbb{Z}$; use $f \in [-\frac{1}{2}, \frac{1}{2})$
- ▶ Sinc interpolation:

$$x_a(t) = \sum_{n=-\infty}^{+\infty} x[n] \frac{\sin(\pi(F_s t - n))}{\pi(F_s t - n)}$$

- ▶ B -bit quantization provides 2^B amplitude levels